INCH-POUND

MIL-T-44214A(GL)
26 January 1989
SUPERSEDING
MIL-T-44214(GL)
31 January 1986

MILITARY SPECIFICATION

T-SHIRT, PHYSICAL FITNESS UNIFORM (PFU)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a cotton polyester blend short sleeve, crew neck t-shirt, used as a component of the Physical Fitness Uniform.

1.2 Classification. The t-shirt shall be of one type in the following sizes (see 6.2).

X-Small (30-32)	Large (42-44)
Small (34-36)	X-Large (46-48)
Medium (38-40)	XX-Large (50-52)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
DDD-T-86	-	Tape, Textile, Cotton: General Purpose (Unbleached, Bleached or Dyed)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage, General Specification For
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon- or Polyester-Covered
MIL-T-43624	-	Thread, Polyester, Spun
MIL-T-44193	-	Thread, Polyester, Textured

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets
MIL-STD-1491	-	Glossary of Knitting Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

AATCC 150-1987-Dimensional Changes in Automatic Home Laundering of Garments

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, North Carolina 27709.)

THE COLOR ASSOCIATION OF THE UNITED STATES, INC.

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

(Application for color cards should be addressed to the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this document may appear in the sample, in which case this specification shall govern.

3.1.2 Standard sample. The finished cloth shall match the standard sample (see 6.4) for shade and appearance. Unless otherwise indicated, the finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 3.3.3.2).

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3.2 First article. When specified (see 6.2), a sample shall be subjected to first article testing and inspection (see 6.3) in accordance with 4.3 and 4.4.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Knitted body cloth. The cloth for the body of the t-shirt shall be jersey knit using a cotton/polyester blend yarn. The cloth shall conform to the requirements listed below when tested as specified in 4.4.1.

Characteristic	Requirement
Weight, ounces per square yard	5.8 ± 0.5
Wales per inch	20 (minimum)
Courses per inch	30 (minimum)
Bursting strength, pounds	75 (minimum)
Fiber content, percent:	
Cotton	50 ± 5
Polyester	Remainder

3.3.2 Cloth, knitted, for collarette. The cloth shall be 1 by 1 rib knit using a cotton/polyester blend yarn. The cloth shall conform to the requirements listed below when tested as specified in 4.4.1.

Characteristic	Requirement
Weight, ounces per square yard	8.25 ± .75
Wales per inch	17 (minimum)
Courses per inch	25 (minimum)
Bursting strength, pounds	50 (minimum)
Fiber content, percent:	
Cotton	50 ± 5
Polyester	Remainder

3.3.3 Color. The color of the jersey knit body and rib knit collarette cloth shall be Heather Gray 467. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyes shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.1.1. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur when tested as specified in 4.4.1.1.

3.3.3.1 Colorfastness. The finished cloth shall show fastness to light, laundering (after three cycles), and perspiration equal to or better than the standard sample, or equal to or better than a rating of "good" when tested as specified in 4.4.1.1. The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.4.1.1.

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3.3.3.2 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.3.4 Nonfibrous material content. The starch and protein content, including chloroform-soluble material of the finished cloth for body collarette shall not exceed 6.0 percent when tested as specified in 4.4.1.1.

3.3.5 Finish. No material having a tendency to cause latent, tendering or yellowing of the cloth shall be used in any of the dyeing or finishing operations. Resin finishes are prohibited.

3.3.5.1 Preshrinking process. The cloth for body and collarette shall be preshrunk by a mechanical shrinking process before being cut into t-shirts such that the t-shirts shall meet the dimensional stability requirements specified in 3.9. The preshrinking process used shall not be identified by name or trade mark on the t-shirt or package.

3.3.5.2 pH. The finished cloth for body and collarette shall have a pH value of not less than 5.0 and not greater than 8.5 when tested as specified in 4.4.1.1.

3.3.6 Thread. The thread for the t-shirts shall be a 50/2 spun polyester conforming to MIL-T-43624 or an equivalent cotton-, rayon-, or polyester-covered staple polyester core conforming to MIL-T-43548, or a high tenacity, textured polyester approximately 45 TEX conforming to type II, class 1 of MIL-T-44193. As an alternate, high tenacity, textured polyester, approximately 24 TEX conforming to type II, class 1 of MIL-T-44193 may be used in specific operations per table I. The color of the thread shall be dyed light gray, DoD Standard Thread Shade AX, C.A. 66064 and shall show good colorfastness to laundering.

3.3.6.1 Thread colorfastness. The thread shall show colorfastness to laundering, perspiration, and light, equal to or better than the standard sample or equal to or better than a rating of "good".

3.3.7 Labels. Each of the t-shirts shall have a combination size, identification and instruction label conforming to type VI, class 14 of DDD-L-20. All required information shall be printed on one side, or the size and identification information may be on one side and care instructions on the other side provided the care instructions are right side up when the label is turned over. All requirements including size of characters shall be as specified in DDD-L-20. The label shall show fastness to laundering. The body of the instruction label shall be as follows:

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CARE INSTRUCTIONS

Hand or machine wash in cold or warm water (maximum 120°F). Do not bleach or iron. Tumble dry at low or medium setting (maximum 160°F).

3.3.8 Tape. The reinforcement tape for the shoulder shall be 3/16 inch wide and conform to Type III, Class 1 of DDD-T-86. As an alternate, a natural white spun polyester tape may be used. The polyester tape shall conform to the following requirements and 3.3.8.1 through 3.3.8.3. Testing shall be as specified in 4.4.1.1:

Characteristic	Requirement
Construction	2 by 2 flat braid
Width	1/8 + 1/16 - 0
Warp, total yarns	21 (min)
Warp, total counts (± 5%)	14/1 or 28/2
Warp strength, lbs.	30 (min)

3.3.8.1 Nonfibrous material. The starch and protein content, including chloroform-soluble and water-soluble material, shall not exceed 4.0 percent.

3.3.8.2 Dimensional stability. The tape shall not shrink more than 4.0 percent in the direction of the warp.

3.3.8.3 Curvature. The tape shall show no more lateral curvature than 1/4 inch within a yard.

3.3.9 Ink. The ink for screen printing "ARMY" to the t-shirt shall be equal to Ruthland Plastic, Inc., product, HO #394, black thermal plastic base ink (withstanding temperatures to 300°F) or equal (see 6.6).

3.4 Design. The t-shirt shall have a crew neck and short sleeves. The collar will be finished with a collarette (see figure 1).

3.5 Construction. The construction shall conform in all respects to the requirements specified in table I and herein. Figure 1 is furnished solely for guidance and information. Should variation from the document appear in figure 1, the document shall govern.

3.5.1 Stitches, seams and stitching. All stitches, seams and stitching types shall conform to FED-STD-751. Seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers, or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching.

3.5.2 Overedge stitching. The gage of the overedge stitching shall be 3/16 ± 1/16 inch. The guide and knives for the overedge machines shall be set to trim only the ravelled edges of the fabric.

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3.5.3 Marking. The black "ARMY" marking shall be screen printed on the front of the t-shirt as shown in Figure 1. Letters shall measure 5/8 ± 1/16 inch in thickness and 3 ± 1/4 inches in height (see figure 2). The overall width shall be 12 ± 1/4 inches and shall be positioned as specified below. Printing shall show colorfastness for a minimum of 15 launderings and dryings when tested as specified in 4.4.1.1.

Size	Distance down from neck seam (± 1/2 inch)	Distance from edge taken from just below the armhole (± 3/4 inch)
X-Small	3	3
Small	3-1/2	4
Medium	4	5
Large	4-1/2	6
X-Large	5	7
XX-Large	5-1/2	8

3.6 Manufacturing operations requirements. The t-shirt shall be made by and with the use of all operations shown in table I. Unless otherwise specified in table I, the contractor is not required to follow the exact sequence of operations listed. Any additional basting or holding stitching used to facilitate manufacture is permissible, providing the thread is removed or does not show on the t-shirt.

3.7 Figures. Figures 1 and 2 are furnished for information purposes only. To the extent of any inconsistencies between this specification and the figure, this specification shall govern.

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NO.	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. All component parts shall be cut lengthwise in the direction of the wales except for the collarette which shall be cut with wales running side to side.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2 shall be removed from production and replaced with non-defective and properly matched components.						
3.	Seam shoulder. Close one shoulder including reinforcement tape in seam.	504	SSa-1	8-12	50/2	50/2	
4.	Attach collarette. Seam double thickness collarette to t-shirt neck opening covering all raw edges and catching the folded label with the size visible at center back. Collarette width shall finish 1 ± 1/8 inch wide.	406	Bsb-1	8-12	50/2 or 24 TEX	50/2 or 24 TEX	
5.	Seam shoulder. Close remaining shoulder seam and collarette including reinforcement tape and tacking collarette end.	504	SSa-1	8-12	50/2	50/2	

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TABLE I. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Sleeves.						
	a. Hem sleeve bottom (s) to finish 1 ± 1/4 inch wide. Hem shall be stitched over the first end of stitching not less than 1 inch.	503	EFc-1	8-12	50/2 or 24 TEX	50/2 or 24 TEX	
	b. Close sleeve along the underarm.	504	SSa-1	8-12	50/2 or 24 TEX	50/2 or 24 TEX	
	c. Set sleeves to armholes.	504	SSa-1	8-12	50/2	50/2 or 24 TEX	
7.	Hem bottom.						
	Hem bottom of t-shirt to finish 1 ± 1/4 inch wide. The stitching shall overlap 1 ± 1/4 inch.	503	EFc-1	8-12	50/2 or 24 TEX	50/2 or 24 TEX	
		or 406	EFa-1	8-12	50/2 or 24 TEX	50/2	
8.	Tacking.						
	The ends of all stitches when not caught in other seams or stitching (except for hem seams and overseaming of collarette) shall be securely tacked by the following method.						
	The first and last ends of stitching chain shall be stitched back into the seam by means of appropriate mechanical device for 1 inch.	1					

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Clean t-shirt. a. Trim and remove any ends of thread of stitching inside and outside of t-shirt. b. Remove all spots and stains.						

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3.8 Finished measurements. Sizes and finished measurements shall conform to the requirements shown in table II. Specific areas to be measured are illustrated in figure 1.

TABLE II. Finished measurements (inches)

Size	Width A/	Length B/	Neck opening C/	Armhole D/	Sleeve under E/	Sleeve over F/	Sleeve opening G/
X-small	18	27	16	8-1/2	3-1/2	6-1/4	5-3/4
Small	20	28	17	9	4	7	6-1/4
Medium	22	29	18	9-1/2	4-1/2	7-3/4	6-3/4
Large	24	30	19	10	5	8-1/2	7-1/4
X-large	26	31	20	10-1/2	5-1/2	9-1/4	7-3/4
XX-large	28	32	21	11	6	10	8-1/4
Tolerance	± 1	± 1	± 1	± 1/2	± 1/2	± 1/2	± 1/2

A/ Width - from side to side with t-shirt laying flat but not stretched, measure across just below the armhole.

B/ Length - from bottom edge to junction of shoulder seam and collarette.

C/ Neck opening - grasp collarette at center back and center front, form a straight line, measure, and multiply by 2.

D/ Armhole - lay t-shirt flat but not stretched and measure from top of sleeve at shoulder to bottom of underarm.

E/ Sleeve under - from bottom of underarm seam to end of arm hem.

F/ Sleeve over - from top of sleeve at shoulder seam to end of arm hem.

G/ Sleeve opening - from folded edge to folded edge when arm hem is laid flat.

3.9 Shrinkage. The completed t-shirts shall not shrink nor elongate more than 10 percent in width or length when tested as specified in 4.4.4.

3.10 Workmanship. The finished t-shirt shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to the requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in table IV, the dimensions specified in table II and tested as specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase specification.

4.4.1.1 Material certification. Unless otherwise specified, a certificate of compliance will be acceptable as evidence that the material characteristics listed in table III conform to the specified requirements when tested by the specified methods.

TABLE III. Material tests

Material	Characteristic	Requirement paragraph	Test method 1/
Jersey knit cloth for body and rib knit cloth for collarette	Type of knit	3.3.1 and 3.3.2	Visual
	Cotton identification	3.3.1 and 3.3.2	1200
	Polyester identification	3.3.1 and 3.3.2	1600
	Weight	3.3.1 and 3.3.2	5041
	Wales and courses per inch	3.3.1 and 3.3.2	5070
	Bursting strength	3.3.1 and 3.3.2	5120
	Fiber content	3.3.1 and 3.3.2	2535
	Labile sulfur	3.3.3	2020
	Colorfastness to:		
	Laundering	3.3.3.1	5614 2/
	Perspiration	3.3.3.1	5680
	Crocking	3.3.3.1	5651
	Light (after 80 standard fading hours)	3.3.3.1	5660
	Nonfibrous material	3.3.4	2611
	pH	3.3.5.2	2811
Screen printing on jersey knit	Colorfastness to:		
	Laundering	3.5.3	5614 3/
Polyester tape	Natural white	3.3.8	Visual
	Spun polyester	3.3.8	----
	Construction	3.3.8	----
	Width	3.3.8	Visual
	Warp, total yarns	3.3.8	5050
	Warp yarn count	3.3.8	4021
	Warp breaking strength	3.3.8	4108 4/
	Nonfibrous material	3.3.8.1	2611
	Dimensional stability	3.3.8.2	5/
	Curvature	3.3.8.3	5/

1/ Specified methods shall be in accordance with FED-STD-191.

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2/ Use three laundering cycles with drying after each cycle.

3/ Use fifteen laundering cycles with drying after each cycle.

4/ Either split drum or flat surface clamps may be used. When flat surface clamps are used, the pulling clamp shall have a uniform speed of 12 ± 0.5 inches per minute, the distance between the jaws shall be 3 inches and the test specimen shall be a minimum of 6 inches.

5/ In accordance with DDD-T-86.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in table IV. The lot size shall be expressed in units of t-shirts. The sample unit shall be one t-shirt. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. The material defects shall be as defined in MIL-STD-1491.

TABLE IV. End item visual defects

Examine	Defect	Classification Major	Minor
Material	Any hole, cut, tear, drop stitch, thin area, or mend.	101	
	Any slub more than twice the thickness of normal yarn.		201
	Any accumulation of slubs, clearly visible, causing nonuniform overall appearance. 1/		202
	Rowey yarn or tiger stripes, loose knitting causing sleaziness, or giving the appearance of shade bars.		203
	Birdseye or miss-knit.		204
	Color not as specified	102	
Shade	Variation in shade within a part or between two or more parts		205
	Streaks clearly visible		206
Thread	Not type, size, or color specified.	103	

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Cleanness	Spot or stain on outside (largest dimension any direction):		
	-More than 1/4 inch up to 1/2 inch inclusive.		207
	-More than 1/2 inch.	104	
	Two or more untrimmed ends more than 1 inch in length inside or outside.		208
Components	Any component missing or other than specified.	105	
Seams and stitching	Any seam twisted, puckered, or pulled, affecting appearance.	106	
	Needle chews.	107	
	Loose tension resulting in a loosely secured seam.		209
	Any part of t-shirt badly pleated, or caught or twisted in any unrelated row of stitching.	108	
	Seams or stitches crack or break when any seam is fully extended.	109	
Seam type	Wrong seam type.	110	
Stitch type	Wrong stitch type.	111	
Stitches per inch	Number of stitches per inch less than minimum or more than maximum specified.		210
Stitching repair	Not as specified	112	
Open seam	Any open seam, except on hems.	113	
	Any open seam on hems 1/2 inch or more.		211

NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur. A double stitched seam is considered open when either side of the seam is open.

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Stitches skipped or broken (stitches skipped or broken on same row of stitching but not amounting to an open seam)	Skipped or broken stitches more than 1/2 inch.	114	
Tacks	Tack omitted on collarette.	115	
	Tack misplaced on collarette not serving purpose intended.	116	
	Tack insecure on collarette.		212
Marking	Any "ARMY" marking omitted, incorrect, illegible, or misplaced.	117	
	Not applied as specified.	118	
	Not size specified.	119	
Colarette	Not double thickness of ribbed cloth.	120	
	Improperly attached to body affecting appearance or serviceability.	121	
	Other than type specified.	122	
	Collarette not uniform in width.		213
	Width less than 7/8 inch or more than 1-1/8 inches.		214
Hem, sleeves and bottom	Omitted.	123	
	Width of hem not uniform throughout.		215
	Width of hem less than 3/4 inch or more than 1-1/4 inches.		216
	Hem not stitched over first end of stitching by at least 3/4 inch		217
Label	Omitted, incorrect, illegible, or misplaced.		218
	Not type or size specified.		219

1/ At normal inspection distance (approximately 3 feet).

4.4.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in table II. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of t-shirts. The sample unit shall be one t-shirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.4 End item shrinkage test. The end item shall be tested for shrinkage as specified in Test Method 150-1987 of AATCC using normal machine cycle, (120° ± 5°F) washing temperature and cotton/sturdy dryer setting (150° ± 10°F). The t-shirt shall not be moistened or pressed before measuring. The measurements shall be taken to the nearest 0.1 inch with the t-shirts laid flat. The t-shirt shall meet the shrinkage requirements specified in 3.9. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of t-shirts. The sample unit shall be one t-shirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of polyethylene bagged unit packs per shipping container is more or less than specified. Number of t-shirts per polyethylene bag is more or less than specified. 1/

1/ For this defect, four polyethylene bagged unit packs shall be examined from each shipping container in the sample.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each t-shirt shall be neatly folded to approximate 11 by 8 inches and inserted in a close-fitting clear polyethylene bag conforming to A-A-50083.

5.1.2 Commercial. The t-shirt shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Ninety-six t-shirts of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The inside dimensions of each shipping container shall approximate 24-1/2 inches in length, 16-1/2 inches in width, and 12 inches in depth for sizes X-Small to Small and 24-1/2 inches in length, 16-1/2 inches in width and 19-1/2 inches in depth for sizes Medium to XX-Large. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Ninety-six t-shirts of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, minimum grade 200 of PPP-B-636. The inside dimensions of each shipping container shall approximate 24-1/2 inches in length, 16-1/2 inches in width, and 12 inches in depth for sizes X-Small to Small and 24-1/2 inches in length, 16-1/2 inches in width and 19-1/2 inches in depth for sizes Medium to XX-Large. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping containers shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III, as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. T-shirts of one size only, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), t-shirts, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

Stock number
Nomenclature
Size
Quantity

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The t-shirts are intended for use by Military Personnel of the Department of the Defense for the physical fitness program.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. When a first article is required (see 3.2, 4.3 and 6.3).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Subject term (key word) listing:

Exercise outfit

6.6 Equal item. Prior to the use of an "or equal item", the supplier shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Preparing activity:

Army - GL

(Project 8415-A651)

Review activities:

Army - MD
DLA - CT

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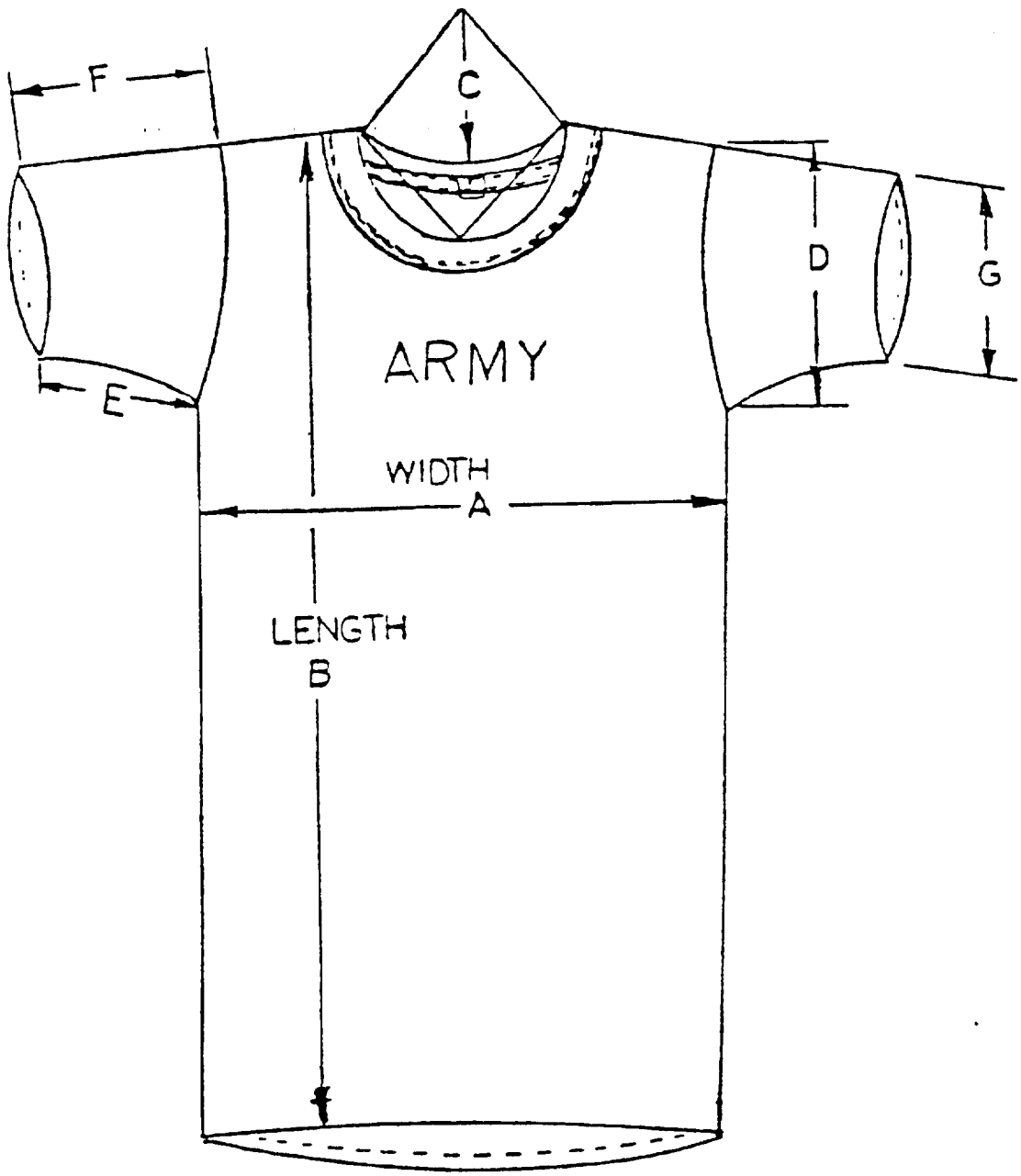


FIG. 1 T-SHIRT
PHYSICAL FITNESS UNIFORM

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2-1-1628



FIGURE 2. THE THICKNESS OF EACH LETTER SHALL BE 5/8 ± 1/16 INCH. THE STYLE OF LETTERING SHALL BE AS SHOWN ABOVE.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-44214A(GL)	T-Shirt, Physical Fitness Uniform (PFU)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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(Fold along this line)

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